

Designed by Remus Teodorescu 2026

LECTURE 11.1

Vision-Based Navigation

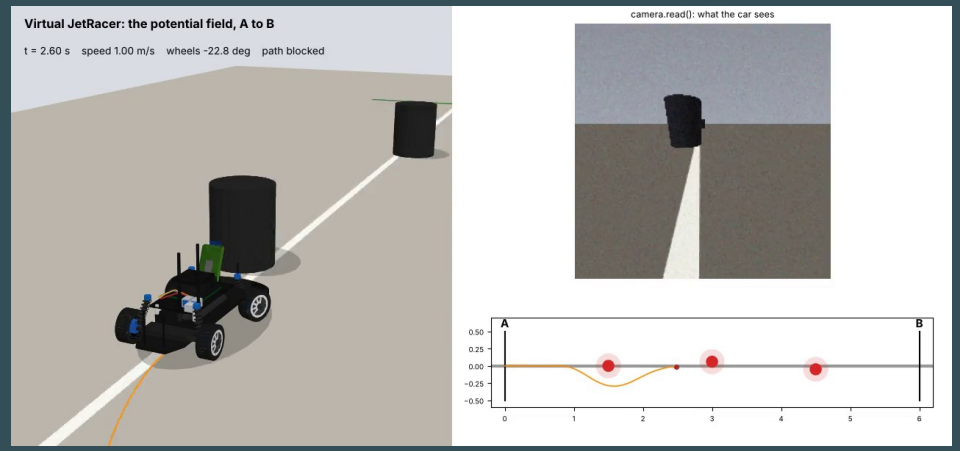
Remus Teodorescu

With support from Research Assistant Noman Khan - nomank@energy.aau.dk

Aalborg University · ret@et.aau.dk

Deep Learning for Engineering · 2026

NEED · RULES · NETWORKS



Cars, warehouse robots, underwater vehicles and humanoid robots steer by what a camera sees, answering two questions: where is the path, and how do I steer onto it? The lecture answers both with fixed rules, then with a convolutional network and the GPU that makes it fast enough.

Learning Objectives

SECTIONS, IN ORDER

1. **The need** — vehicles and robots that steer by camera
2. **Conventional vision** — fixed rules, and where they fail
3. **Deep learning** — a network that steers, its data, its GPU

Two questions run through the lecture: where is the path, and how do I steer onto it? Each is answered first in words, then in mathematics - by fixed rules, then by a network - on Ex_11's virtual JetRacer, where the true position of the line is known.

READ, AND WHAT EACH ANSWERS

UDL ch. 1	where is deep learning applied?
UDL ch. 10	why do images need convolutions?
UDL ch. 8	what must the training data show?

Recap of L10.2 Electrolysers

1 – 2 · A NETWORK FOR THE FUEL CHANNEL

1 **How can one network learn the steam fraction, temperature and current density along the channel at every operating point?**

The network takes the position along the channel, the cell voltage and the inlet temperature, and returns the three fields. It is trained over 1.15 to 1.35 volts and 750 to 850 degrees until the three laws hold at points over the channel and the operating range; the inlet values are built into the form of the solution, and the current is kept below the limiting current. Any operating point is then read off, not solved again. *Liu ch. 2.2; Ex_10.2 section 5*

2 **Why does the local current make the channel harder to solve, and how is the network's answer checked?**

The current has no equation of its own along the channel: at every point it is the root of the charge transfer at the local steam fraction and temperature, so no field can be solved first. The network takes it as a third residual, in units of 50 millivolts so it weighs like the balances. Scored against a boundary-value solver at twelve operating points, it is within 0.0022 of the steam fraction and 0.19 kelvin, three times closer than finite volumes on 50 cells. *Ex_10.2 sections 3 – 5*

3 – 4 · THE OPTIMAL VOLTAGE

3 **How can the optimal cell voltage be found with a network, and what does the gradient save?**

The cost of an hour, the electricity bought less the hydrogen sold plus the share of the stack aged, is read from the network's fields, with the 10 kelvin limit as a penalty. Autograd through the network gives its derivative in the voltage, and gradient descent finds 1.250 volts in seconds; without the limit, 1.167 volts, where the cell cools 25 kelvin. A sweep of 81 solver runs agrees. The prices and the ageing law are assumptions, so the optimum is conditional on them. *Liu ch. 5; Ex_10.2 section 6*

4 **Why does a network help to optimise the operation of the cell over its lifetime, and what should not be claimed from it?**

Over a lifetime the decision is a trajectory of voltage and temperature: what produces more now ages the cell faster. A solver gives the cost, but its gradient costs one solve per variable per step, or an adjoint built by hand; a trained network gives every gradient by automatic differentiation. The degradation constants here are assumed, and published rates span an order of magnitude, so a predicted lifetime ranks schedules; it is not a number to promise. *Liu ch. 5*

Autonomous Vehicles and Robots

Vision-based navigation — steering a vehicle or a robot from what its camera sees: where the free path is, where it leads, and what is in the way. The camera is cheap and carries the most information of any sensor, but turning its pixels into a decision is the hard part.

FOUR OF THEM, AND THE LOOP THEY SHARE



robotaxi



warehouse AGV



humanoid



underwater vehicle

see one frame → **decide** path, obstacles → **act** steer, speed → **in time** fast enough

Photographs: Wikimedia Commons. This lecture is the path; L11.2 is what blocks it.

WHAT MAKES THE CAMERA'S JOB HARD

	specific challenges	beside the camera
road car	night, rain, low sun; 3.6 m blind per 0.1 s	radar, LiDAR, GNSS
robotaxi (Cybercab)	no wheel or pedals: no driver to take over	none: cameras only
warehouse AGV	shiny floors, narrow aisles, people and forklifts	safety LiDAR, floor markers
underwater vehicle	light gone within metres; murky water	sonar, Doppler log, inertial
humanoid robot	camera sways; stairs need depth	depth camera, joint angles

The JetRacer Pro

JetRacer Pro — a small car that computes on board: a Jetson Nano runs the network on its GPU, a wide camera is its only sensor, a servo steers and a brushed motor drives. It is the scaled-down case of the machines on the previous slide, with the same loop: see, decide, act.

WHERE A ROW OF PIXELS LANDS ON THE FLOOR

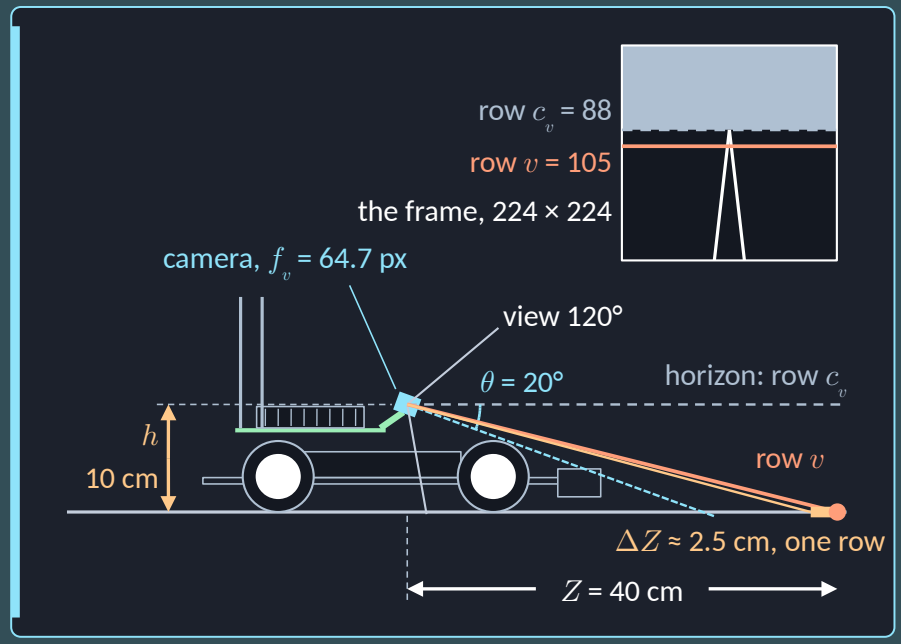
$$v - c_v = \frac{f_v\, h}{Z}, \qquad \Delta Z \approx \frac{Z^2}{f_v\, h}$$

v the image row, c_v the horizon's row, f_v the focal length in pixels
h the camera's height, Z the distance ahead, ΔZ the floor one row covers

THE VERIFIED SPECIFICATION

compute	Jetson Nano 4 GB, 128-core Maxwell GPU
camera	IMX219-160: 8 MP, 160° lens; the network reads 224 × 224
steering	Ackermann, 6 kg·cm servo; 4WD, RC380 brushed motor
pack	8.4 V, four 18650 cells, 2S2P; no encoder

THE CAMERA ON THE CAR, SIDE VIEW



To scale. The camera's height, tilt and field of view are the virtual JetRacer's, assumed; they are measured on the car.

Vision by Hand-Written Rules

Conventional vision — fixed steps chosen by an engineer. The white tape reflects more light than the dark floor, so pixels brighter than the floor mark the line; the centre of the bright run is where the tape crosses the row - the point to steer toward. Nothing is learned from data.

THE DETECTOR, AS EX_11 WRITES IT

$$w_j = \max\left(I_j - \bar{I} - 0.15,\ 0\right), \qquad \hat{x} = \frac{\sum_j w_j x_j}{\sum_j w_j}$$

I_j the brightness of pixel j, $\bar{I}$ the floor's median, x_j its column
w_j its weight, $\hat{x}$ the estimated column of the line

FROM A FRAME TO A COMMAND

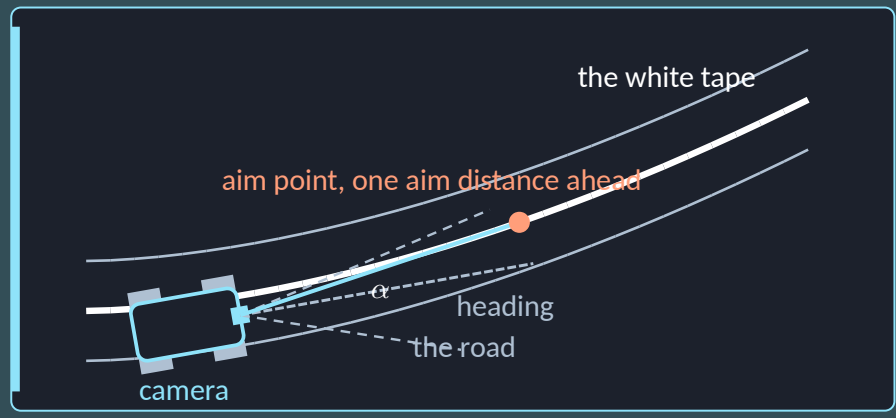
frame grey pixels → **threshold** set by hand → **line** bright centre → **rule** steer to it

every step has a number chosen by an engineer, for one floor and one light

THE STEERING RULE

$$\delta = \arctan\frac{2L\sin\alpha}{\ell_d}$$

δ the steering angle, α the angle to the aim point
L the wheelbase, 0.20 m, ℓ_d the aim distance, 0.40 m



The rule is pure pursuit: steer onto the circle through the aim point; L and ℓ_d assumed until measured.

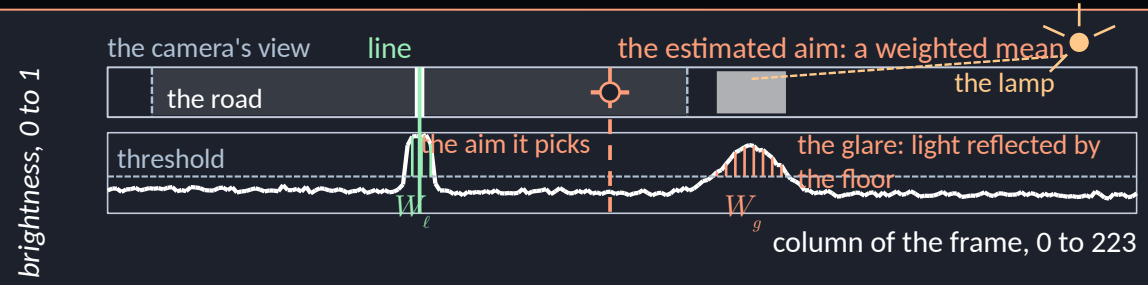
Where Conventional Vision Fails

Robustness — keeping the error within what was agreed when the scene changes: the light, a reflection, a shadow, a new floor, an unseen object. A hand-tuned rule is robust only to the changes its designer thought of.

WHY THE ESTIMATED AIM MOVES TOWARD THE GLARE

$$\hat{x} = \frac{W_\ell x_\ell + W_g x_g}{W_\ell + W_g}$$

W_ℓ, x_ℓ, W_g, x_g the summed weight and mean column of the bright pixels at the tape and at the glare
$W = \sum_i w_j$ over each group, with $w_j = \max(I_j - \bar{I} - 0.15,\ 0)$ the weight of one pixel; the labels ℓ, g are ours

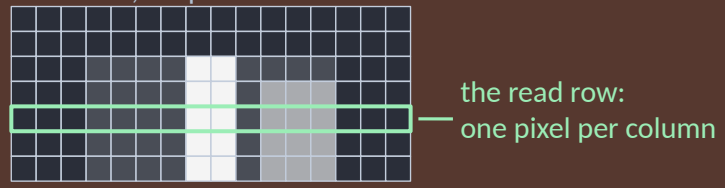


TALK · 2 MIN How would you explain, in your own words, why a rule that works on one floor can fail on another? Then, in the plot above, why does the estimate land between the line and the glare?

THE DETECTOR ON 300 FRAMES

the scene	mean $\lvert\hat{x} - x\rvert$, the frame = -1 to +1
along the line	0.007
beside the line	0.021
with glare	0.083
the agreed error	0.03

the frame, simplified



The curve on the left reads the green row, one pixel per column. Each number is the mean of $\lvert\hat{x} - x\rvert$ over a scene's 300 test frames; 0.03 is about 3 pixels of the 224.

Convolutional Network

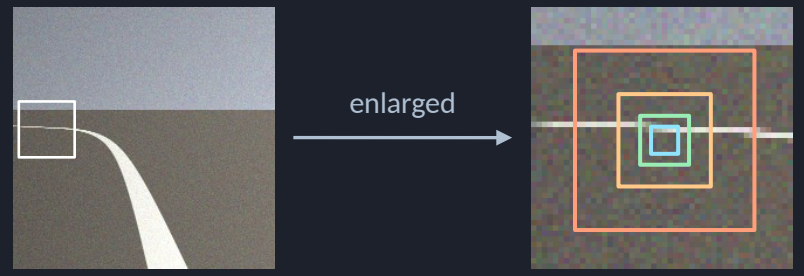
Convolutional network — a network whose layers slide small kernels, a few pixels wide, over the whole frame. Each kernel takes a weighted sum of the pixels under it, so it finds the same pattern wherever it lies. The weights are learned from labelled frames, not set by hand, and each layer combines the one before over a wider window.

RULES SET BY HAND AGAINST LEARNED KERNELS

	conventional vision	convolutional network
the feature	brighter than the floor	learned from labelled frames
the kernel	set by hand, e.g. Sobel	learned, 8 to 32 per layer
what it reads	one band of rows	the whole frame
it fails on	what was not foreseen	what was not shown

In L5.1 the same idea sorted weld radiographs into clean, crack and pit: a kernel set by hand found the edge of the crack, and a small network learned its own kernels. Here the picture is the floor, and the answer is where the line is.

WHAT EACH LAYER SEES



layer	output	sees	weights
1, 5 × 5	112 × 112 × 8	5 px	608
2, 3 × 3	56 × 56 × 16	9 px	1 168
3, 3 × 3	28 × 28 × 32	17 px	4 640
4, 3 × 3	14 × 14 × 32	33 px	9 248
dense	32	all	200 736
output	1 number	all	33

216 433 weights, 93 % of them in the dense layer.

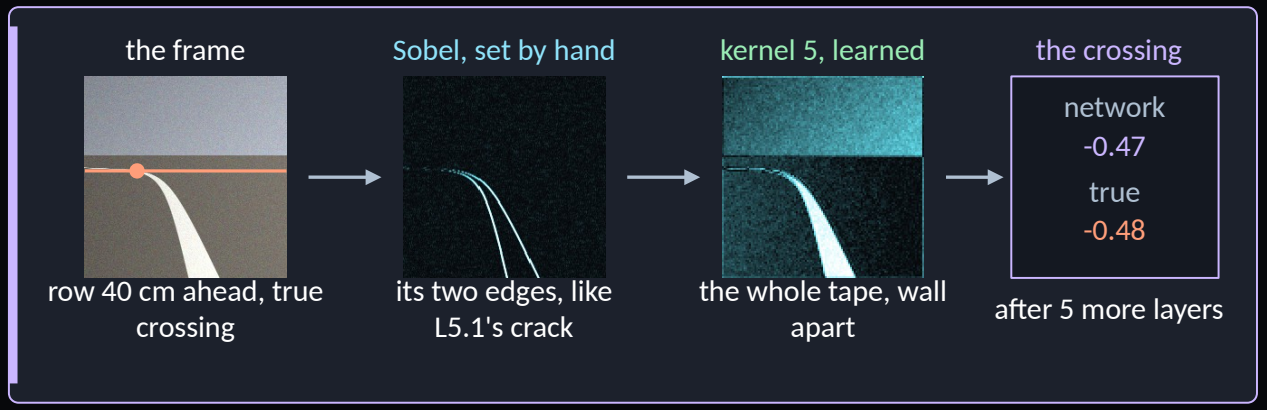
Steering From a Frame

Steering network — a convolutional network that reads one camera frame and returns one number: where the taped line crosses the image row that sees the floor 40 cm ahead, from −1 at the left edge to +1 at the right. The car steers by that number times a gain. Its kernels are learned from frames whose true crossing is known.

ONE CONVOLUTION, AS IN L5.1

$$y_{ij} = \sigma\Big(\sum_{m,n} w_{mn}\, x_{i+m,\, j+n} + b\Big)$$

x the frame, w the kernel, b its bias, σ the ReLU, y the feature map



DISTANCE FROM THE TRUE CROSSING

300 test frames	detector	network
on the line	0.8 cm	1.8 cm
beside the line	4.5 cm	3.6 cm
with glare	14.1 cm	5.3 cm
largest miss, glare	79 cm	32 cm

Root mean square, across the floor 40 cm ahead; lower is better. The truth is exact: the twin draws each frame from a known pose and computes where the tape crosses that row. On the line the hand-set detector is closer; beside it and with glare the network is. Its price: 80 s of training and twice the time per frame.

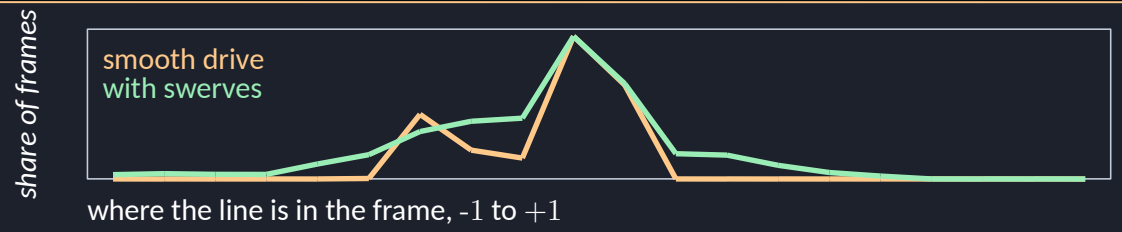
Learning From a Human Driver

Training drive — a person drives the car round the course with a gamepad while the camera saves a frame ten times a second, each labelled with where the line truly is. Like the survey cars that drive every road once to build a map, the drive is done once, before training. The network learns only what this drive shows.

THE LOSS IS AN AVERAGE OVER THE DRIVE'S FRAMES

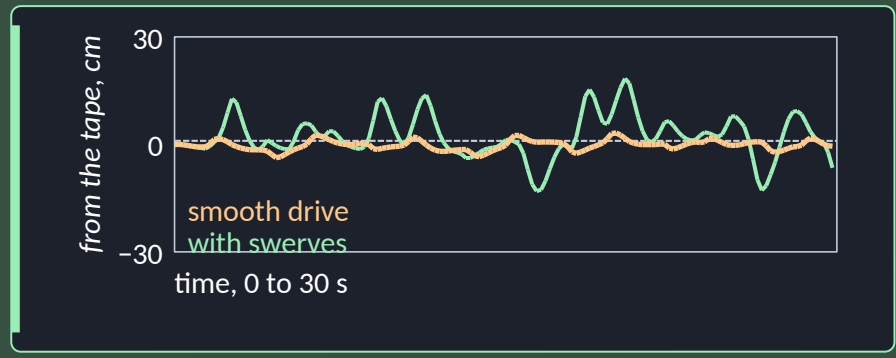
$$\mathcal{L} = \frac{1}{N}\sum_{k=1}^{N}(\hat{y}_k - y_k)^2$$

N the frames of the drive, 1500; $\hat{y}_k$ the network's crossing in frame k, y_k the true one



A driver who never leaves the line shows the network no way back to it.

TWO TRAINING DRIVES



mean error, test frames	smooth	swerves
car on the line	0.010	0.021
car beside the line	0.195	0.036
with glare	0.194	0.050

Trained on the smooth drive the network is 5 times worse beside the line, where a drifting car needs it. Same network, same 1500 frames: only the drive differs.

Why a GPU

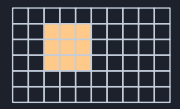
GPU — a processor with many small cores that run the same operation on different data at the same time. Every output of a convolution is its own sum of products over a small window, independent of the others, so a GPU computes them all at once.

THE DISTANCE DRIVEN WITHOUT NEW INFORMATION

$$d = \frac{v}{f} = v\,(t_{\text{cap}} + t_{\text{pre}} + t_{\text{net}} + t_{\text{act}})$$

d the blind distance, v the speed, f decisions per second
t the times to capture, preprocess, run the network and act

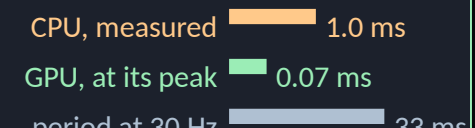
THE SAME CONVOLUTION ON A CPU AND ON A GPU



CPU: one window after another



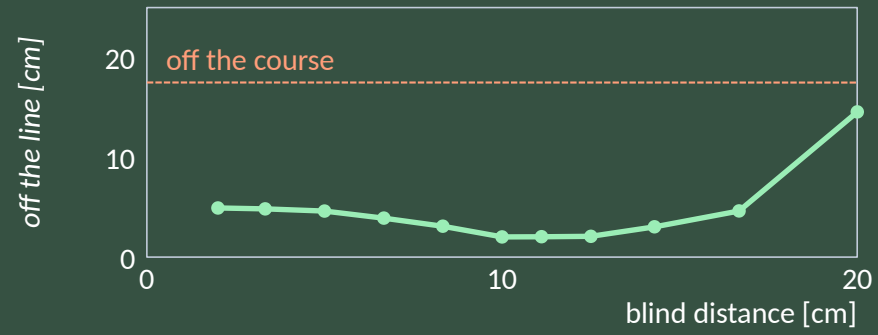
GPU: all windows at once



time for one frame, log scale

TALK · 2 MIN How would you explain, in your own words, why a slightly more accurate but slower network can be a worse driver? Then, in the plot on the right, where does the car start to swing wide?

THE LAP, AND THE WORK PER FRAME



$$\text{MAC} = H_{\text{out}}\, W_{\text{out}}\, C_{\text{out}}\, C_{\text{in}}\, k^2$$

$H_{\text{out}}, W_{\text{out}}$ the output size, $C_{\text{in}}, C_{\text{out}}$ the channels, k the kernel

one MAC	a multiply and an add = 2 FLOP
Ex_11's network	17 million MAC a frame
laptop CPU, measured	0.9 ms a frame
Nano GPU, at its peak	0.07 ms: 236 billion MAC/s

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
semantic segmentation	which pixels are free to drive on	UDL ch. 10	floors with no line
depth from one camera	distance from a single frame	UDL ch. 10	no range sensor on board
vision transformers	attention over patches of the frame	UDL ch. 12	large, varied data sets
imitation learning	a policy from human driving	UDL ch. 19	road vehicles
reinforcement learning	a policy from reward, by trial	UDL ch. 19	simulators, legged robots

WHY THEY ARE LEFT OUT

Ex_11 follows one taped line with one camera, scored on rendered frames where the true position is known, so none of them is needed to do it. They matter for the full-size machines of the first section.

EXERCISE

Exercise 11

ONE NOTEBOOK · THIS LECTURE: SECTIONS 1 – 4 AND 7

- **1 – 3** — the virtual JetRacer, its frames, the detector
- **4** — a network that steers; TODO 1, the convolutions
- **7** — the loop rate and the GPU, this lecture's part
- **5 – 6, fewer bits** — left for L11.2: obstacles and strategies
- **8 – 11** — the race, the report and the mini projects



the same task on the real car: a demonstration in the lab

THE QUESTION EACH PART ANSWERS

1 – 3	when is a fixed rule enough?
4	what must the training drive show?
7	how fast must the loop be?
for L11.2	sections 5 – 6, and the report

REPORT QUESTION, TO CONCLUDE

Which avoidance strategy reaches B fastest without a collision, and what limits its speed: the errors of the networks, the loop rate, or the distance needed to react?

SUMMARY

Key Takeaways

VISION POLICY

what it is	a CNN: frame to where the line is
function	where the line is, a little ahead
train, tune	a human drive that swerves and back
vs a fixed rule	no hand tuning, holds under glare
used for	steering cars, robots, vehicles

THE LOOP ON A GPU

what it is	many cores, one operation each
function	all outputs of a convolution at once
train, tune	the whole loop, measured on the car
vs accuracy	judged together, rate first
used for	a short blind distance $d = v/f$

Next: what blocks the path - obstacles, and what to do about them.

Questions

1 – 2 · A NETWORK THAT STEERS

1 **Why can a convolutional network steer from a camera frame where a hand-tuned line detector fails?**

The detector follows fixed steps chosen by an engineer: it keeps the pixels brighter than the floor in the row it reads and steers to their mean column, so anything bright pulls the aim toward it. The network learns its own features from frames whose true position of the line is known. On clean frames the detector meets the agreed error; with a patch of glare its error rises to 0.08, while the network, shown glare in training, keeps it at 0.05. *UDL ch. 10; Ex_11 sections 3 – 4*

2 **What must the training frames show for the network to steer a car that is drifting off the line?**

The network learns only what the frames show: the floor, the camera mount and the light it will drive in, the disturbances such as glare, and the positions the car reaches when it is already going wrong, labelled with the way back. So the person who drives the car to record them also swerves away from the line on purpose. Trained on a smooth drive, the same network is about five times worse beside the line, exactly where a drifting car needs it. *UDL ch. 8; Ex_11 section 4*

3 – 4 · THE LOOP ON A GPU

3 **Why is a steering network judged by the rate of the control loop as well as by its error?**

Between two decisions the car drives on its last command with no new information, over a blind distance of its speed divided by the decisions per second. With the network steering the virtual car round the lap it stays within 5 cm up to 17 cm blind and swings 15 cm at 20 cm, so a slightly more accurate network that halves the loop rate is usually a worse car. The whole loop is timed, from capture to command, because the network is only one of its stages. *Ex_11 section 7*

4 **Why does a convolutional network need a GPU to steer in time, and what must be measured on the car?**

Every output of a convolution is its own sum of products over a small window, independent of the others, so a GPU's many cores compute them all at once; the work is counted in MAC, a multiply and an add. The exercise's network takes a millisecond on a laptop processor, inside the 33 ms frame period; a ResNet-18, the real car's network, does a hundred times the work and fits only on the GPU. The loop rate itself is measured on the car, from capture to command. *UDL ch. 10; Ex_11 section 7*

NEXT

L11.2 — Obstacle Avoidance

What blocks the path: a network that tells free from blocked, the strategies that steer around an obstacle, and the distance the car needs to react.

BEFORE THE NEXT SESSION

read	UDL ch. 10
run	Ex_11's notebook, sections 1 to 4
bring	your detector and network errors
think	what in your frames is an obstacle?

The real JetRacer drives the same task in a lab demonstration, with the throttle limited in software and one hand on the stop.